

## **Electronic Crime Scene Investigation**

### **1.0 Introduction**

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**1.01: What is Electronic Evidence:** Electronic Evidence is information and data of investigative value that is stored on or transmitted by and electronic device. Such evidence is acquired when data or physical items are collected and stored for examination purposes.

#### **1.01(a): Electronic Evidence:**

- **Is often latent in the same sense as fingerprints or DNA evidence is latent.**
- **Can transcend borders with ease and speed.**
- **Is fragile and can be easily altered, damaged, or destroyed.**
- **Can be time-sensitive.**

**1.02: How is Electronic Evidence Handled at the Crime and/or Incident Scene:** Precautions must be taken in the collection, preservation and examination of Electronic Evidence.

**1.02(a): Handling Electronic Evidence normally consists of the following steps:**

- **Recognition and identification of the evidence.**
- **Documentation of the crime and/or incident scene.**
- **Collection and preservation of the evidence.**
- **Packaging and transportation of the evidence.**

**1.03: Is Your Agency or Organization Prepared to Handle Electronic Evidence?** It is highly recommended that your Agency or Organization have the procedural and technical capabilities in place and that they be thoroughly trained and tested prior to attempting to secure Electronic Evidence. It is recommended that procedural plans (e.g. Computer Security Incident Response Plan) be developed in compliance with departmental policy and that they adhere to all Federal, State and Local Laws.

**1.04: Electronic Devices: Types and Potential Evidence:** Electronic Evidence can be found in many of the new type of electronic devices available to today's consumers. The following are but some of the places that Electronic Evidence may be located:

**1.04(a): Central Processing Unit:**

- **Desktop**
- **Laptop**
- **Tower**
- **Modular Rack Mounted**
- **Minicomputer**
- **Mainframe**

**1.04(b): Access Control Devices:**

- **Smart Cards**
- **Dongles,**
- **Biometric Scanners**

**1.04(c): Telephone Switching Systems (PBX)**

**1.04(d): Answering Machines**

**1.04(e): Facsimile Machines**

**1.04(f): Printers**

**1.04(g): Personal Digital Assistants**

**1.04(h): Memory Cards**

**1.04(i): Network Components (Routers, Hubs, Switches)**

**1.04(j): Pagers**

**1.04(k): Removable Storage Drives:**

- Thumb Drives (USB)
- Jazz Drives
- Zip Drives
- DAT Tape
- Floppy Disk
- DLT Tape Cartridge
- Removable Hard Drives
- Etc.

**1.04(l): Miscellaneous Electronic Storage Items:**

- Caller Identification Device
- Copiers
- Credit Card Skimmers
- Digital Watches
- Global Positioning System Terminals

**1.05: Investigative Tools and Equipment: Common and very specialized equipment will be required in order to properly secure Electronic Evidence. Example:**

**1.05(a): Tool Kit**

- Documentation Tools:
  - o Cable Tags
  - o Indelible Felt Tip Markers
  - o Stick On Labels
- Disassembly and Removal Tools:
  - o Flat Blade and Philips type screwdrivers
  - o Hex Nut Drivers
  - o Needle Nose Pliers
  - o Secure Bit Drivers
  - o Small Tweezers.
  - o Specialized Screwdrivers (manufacture specific (e.g. Compaq, Macintosh))

- **Package and Transport Supplies:**

- o Antistatic Bags
- o Antistatic Bubble Wrap
- o Cable Ties
- o Evidence Bags
- o Evidence Tape
- o Packing Materials
- o Packing Tape
- o Sturdy Boxes
- o Gloves
- o Hand Truck
- o Large Rubber Bands
- o Etc.

- **Specialized Investigative Software**

- o Encase
- o Forensic Tool Kit
- o ProDiscover
- o Etc.

**1.06: Securing and Evaluating The Scene:** The first responder should take steps to ensure the safety of all persons at the scene and to protect the integrity of all evidence, both traditional and electronic.

- **Secure and Evaluate The Scene:**

- o Removal of non-essential personal from the immediate area.
- o Information System State: IF OFF, LEAVE IT OFF, IF ON LEAVE IT ON.
- o Protect perishable data, both physically and electronically.
- o Identify and document all connections, both data and electrical.
- o Conduct preliminary interviews.

**1.07: Documenting The Scene:** Documentation creates a permanent historical record of the scene. Documentation is an ongoing process throughout the investigation. It is important to accurately record the location and condition of the computers, storage media, other electronic devices and conventional evidence.

**1.08: Electronic Evidence Collection:** Computer evidence must be handled carefully and in a manner that preserves its evidentiary value. This relates not just to the physical integrity of an item or device, but also to the

electronic data it contains. Certain types of computer evidence, therefore, require special collection, packaging and transportation. Consideration should be given to protect data that may be susceptible to damage or alternation from electro-magnetic fields (e.g. static electricity, magnets, radio transmitters, etc).

**1.09: Packaging, Transportation and Storage: Actions taken should not add, modify or destroy data stored on a computer or other media.**

**1.10: Forensic Examination by Crime Category:**

- Auction Fraud
- Child Exploitation/Abuse
- Computer Intrusion
- Death Investigation
- Domestic Violence
- Economic Fraud
- E-Mail Threats/Harassment/Stalking
- Extortion
- Gambling
- Identify Theft
- Narcotics
- Prostitution
- Software Piracy
- Telecommunications Fraud

**1.11: Searching and Seizing Computers Without A Warrant: Extensive discussion concerning the Fourth Amendment to the U.S. Constitution's re; An Individual Reasonable Expectation of Privacy.**

**1.12: Searching and Seizing a Computer With A Warrant: Again, extensive discussion on the procedures and requirements for executing a search.**

**1.13: Electronic Surveillance In Communication Networks: Extensive discussion concerning requirements and capabilities for monitoring an individuals specific acts in a dedicated communications network.**

**1.14: Information System Security Incident Response Planning: Practical application for development of a STRAWMAN Incident Response Plan for use in team development at the corporate level. Exercise would include, but not be limited to:**

- Establishment of a Team
- Development of Team Charter

- Mission
- Scope
- Organizational Structure
- Information Flow
- Services Provided
- Management & Leadership
- Establishment of Advisory Board
- Identification of Permanent Team Members
- Identification of Temporary Team Members
- Incident Severity and Declaration
- Incident Severity
  - Level One through Level Five
- Incident Declaration
- Response Procedures
- Triage Phase
- Response Phase
- Recovery Phase
- Maintenance Phase

“ Every company or organization needs to have a Information System Security Incident Response Plan in place, regardless of where the company is in the security spectrum. Security incidents won’t wait for new security software to be installed, or security consulting engagements to be completed. A company has to be prepared to defend what is has and to response to security incidents as they occur. - Commercial security incident response services can help companies develop their CRISP’s not only with consulting to build the plan itself, but also by providing response team personnel. A company can augment its own security staff whatever its size and experience with a commercial service to provide additional expertise and specialized skills. This partnership approach results in the best protection the lowest cost. “ - **Security Incident Response Systems, 2002**

*Source: Searching and Seizing Computers and Obtaining Electronic Evidence in Criminal Investigations – Computer Crime and Intellectual Property Section, Criminal Division, U.S. Department of Justice, dtd July 2002*

*Source: Electronic Crime Scene Investigation – A Guide For First Responders, U.S. Department of Justice, Office of Justice Programs, National Institute of Justice, dtd 2001.*

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